Finite Element Modeling & Geometric Optimization for Ultrasonic Fatigue & In-Situ Analysis of Sintered High-Entropy Alloys

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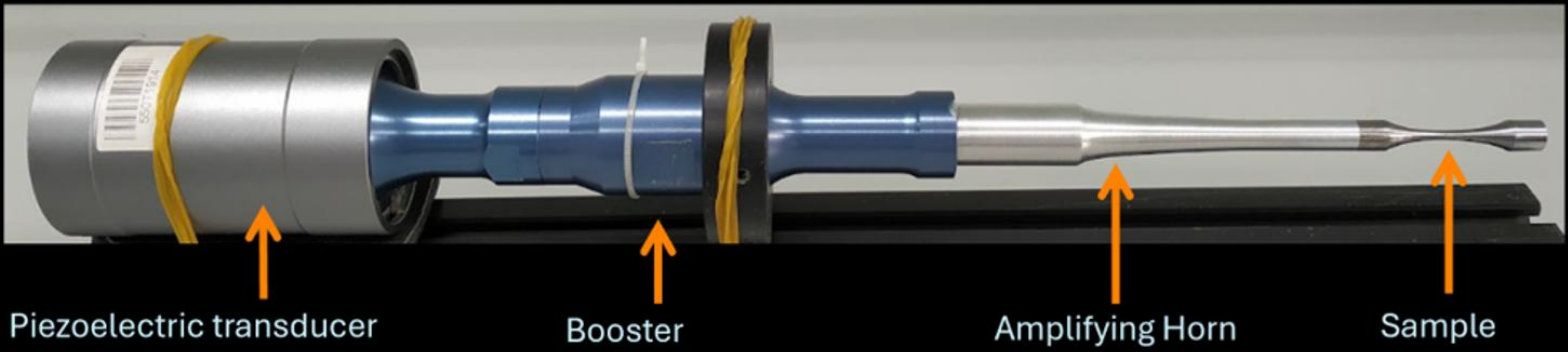
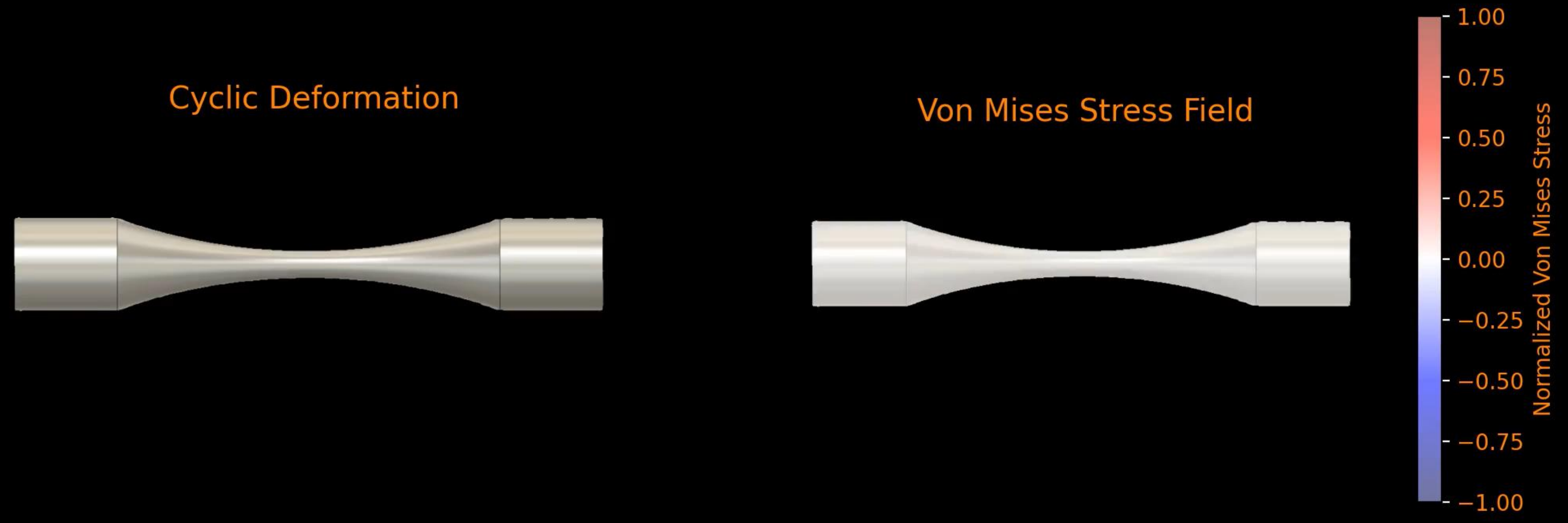
UT-ORII SMaRT Symposium 2026

Finite Element Modeling & Geometric Optimization for Ultrasonic Fatigue

What is Ultrasonic Fatigue?





Why is This Important?

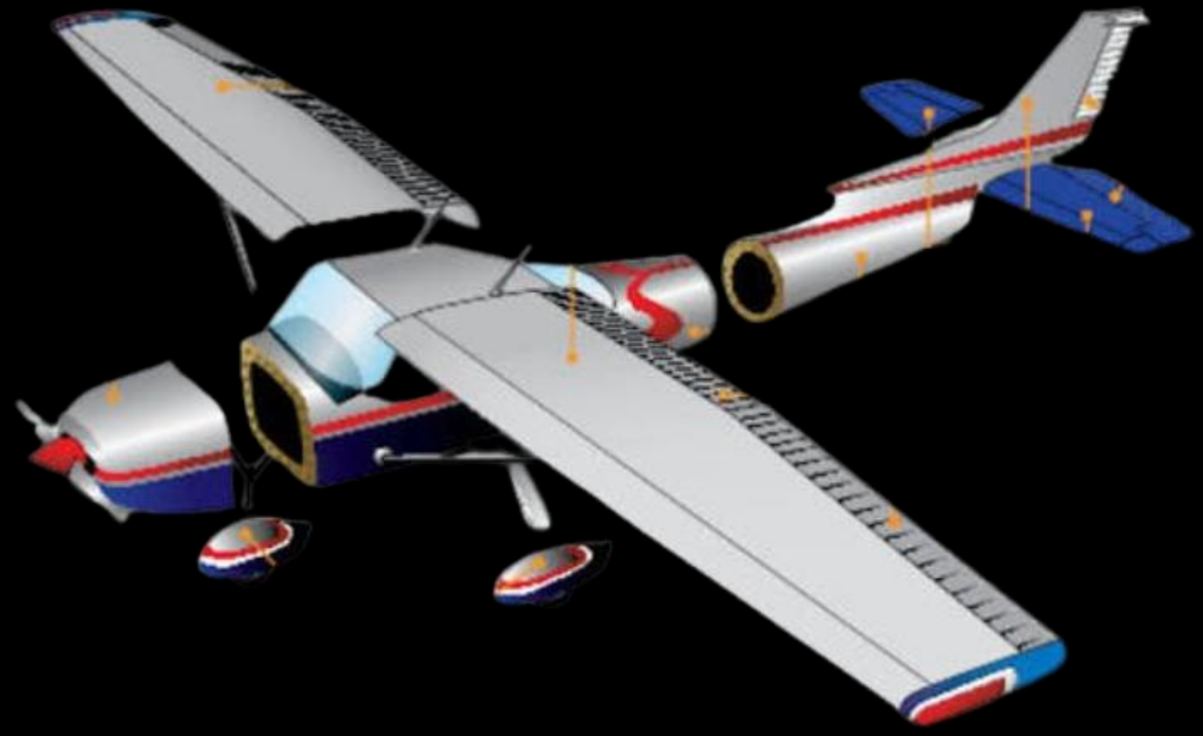


Image by HelloArtsy

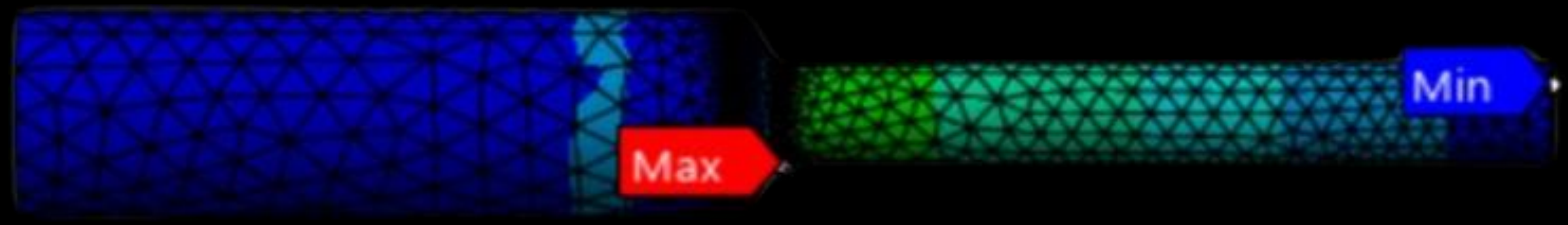
Image by Moog

Image by Railway Age

What Did I Do?

Original Design → Select Material → Establish Design Requirement → Modal and Harmonic Analysis → Optimize Geometry

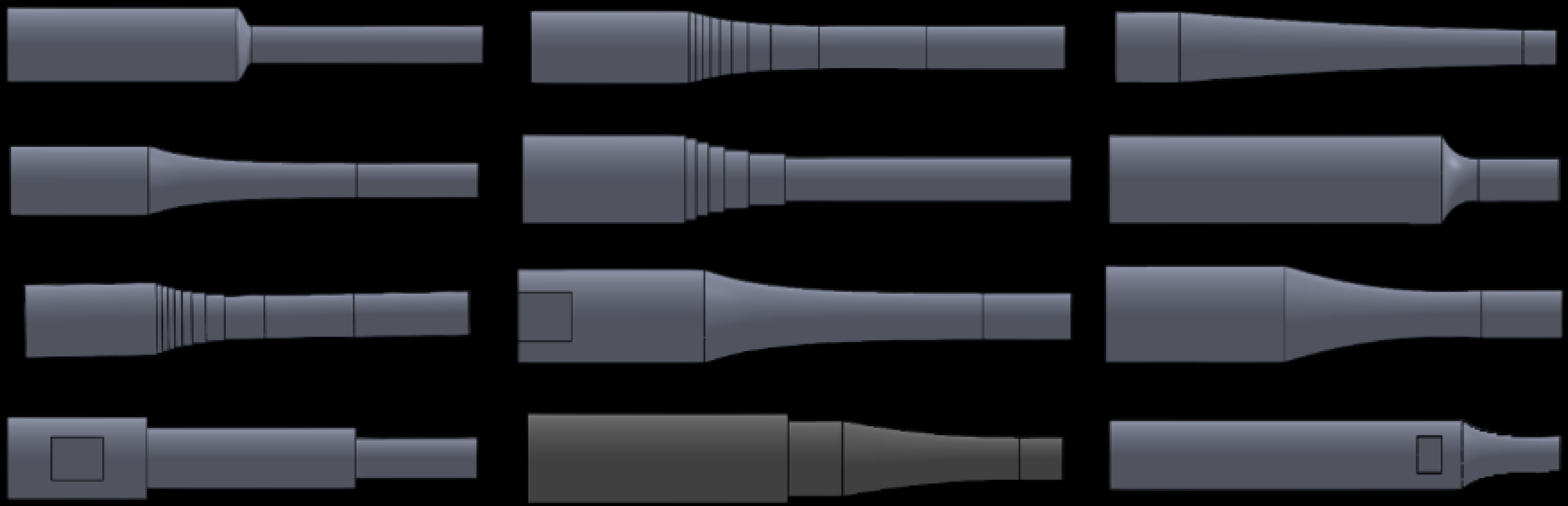
18967 Hz, 2.762x Gain



20000 Hz, 3.956x Gain

Through countless iterations, this design had the highest gain, while maintaining 20000 Hz

Horn Designs



And many more…

Results

Hourglass Dimensions Before & After

Hourglass	Original Length	Optimized Length	Original Resonance	Optimized Resonance
4140 Carbon Steel	61.60 mm	61.60 mm	19959 Hz	19959 Hz
Stainless Steel 316	61.60 mm	61.08 mm	19780 Hz	19957 Hz
Al 6063-T52	61.60 mm	61.08 mm	19799 Hz	19976 Hz
Al 2024-T4	61.60 mm	61.08 mm	19805 Hz	19983 Hz

Amplifying Horn Original Dimensions

Amplifying Horn	Large Cylinder Length	Small Cylinder Length	Curve Axial Length	Horn Type
Al 6061 v1	40.005 mm	35.052 mm	59.944 mm	Exponential Horn
Ti 64 v1	40.000 mm	25.000 mm	60.000 mm	Exponential Horn
Ti 64 v2	65.532 mm	60.960 mm	8.890 mm	Exponential Horn

Amplifying Horn Optimized Dimensions

Amplifying Horn	Large Cylinder Length	Small Cylinder Length	Curve Axial Length	Horn Type
Al 6061 v1a	63.410 mm	65.240 mm	4.000 mm	Cosine Horn
Al 6061 v1b	55.730 mm	25.200 mm	61.280 mm	Conical Horn
Ti 64 v1a	61.990 mm	63.780 mm	4.000 mm	Cosine Horn
Ti 64 v1b	55.540 mm	22.920 mm	61.280 mm	Conical Horn
Ti 64 v2a	62.450 mm	63.000 mm	4.000 mm	Cosine Horn

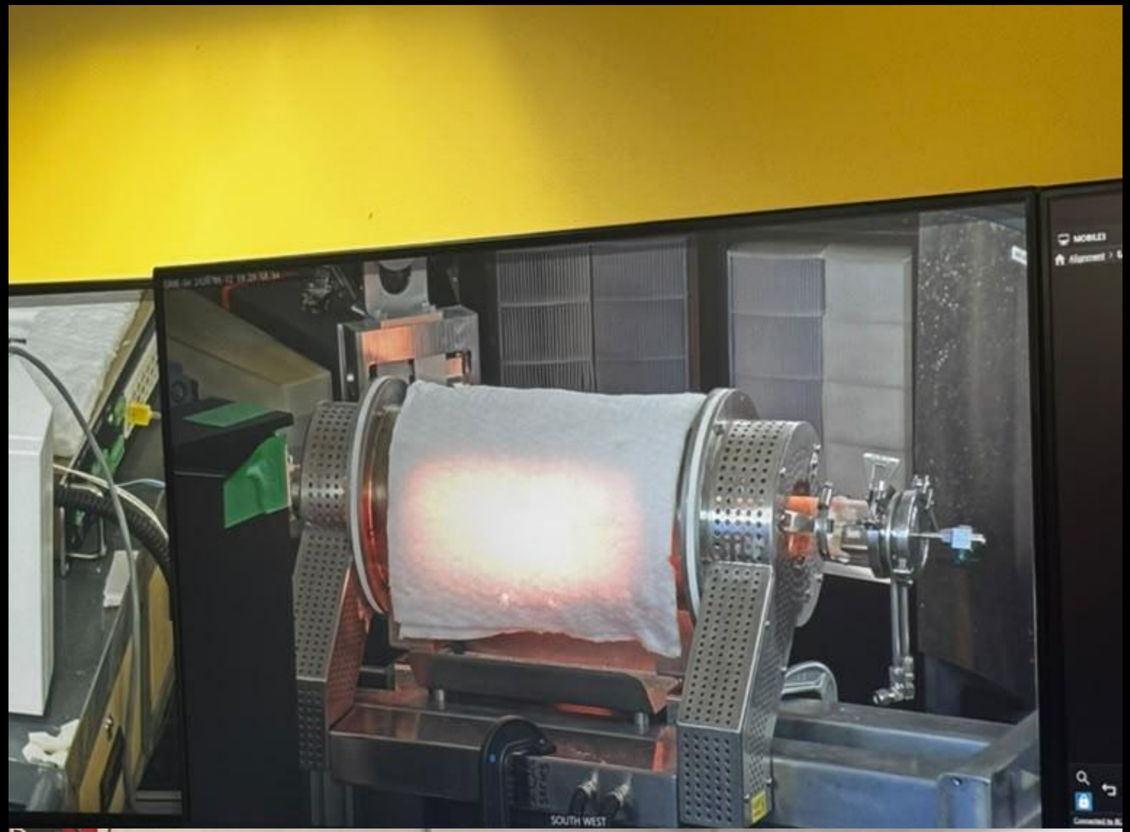
Findings

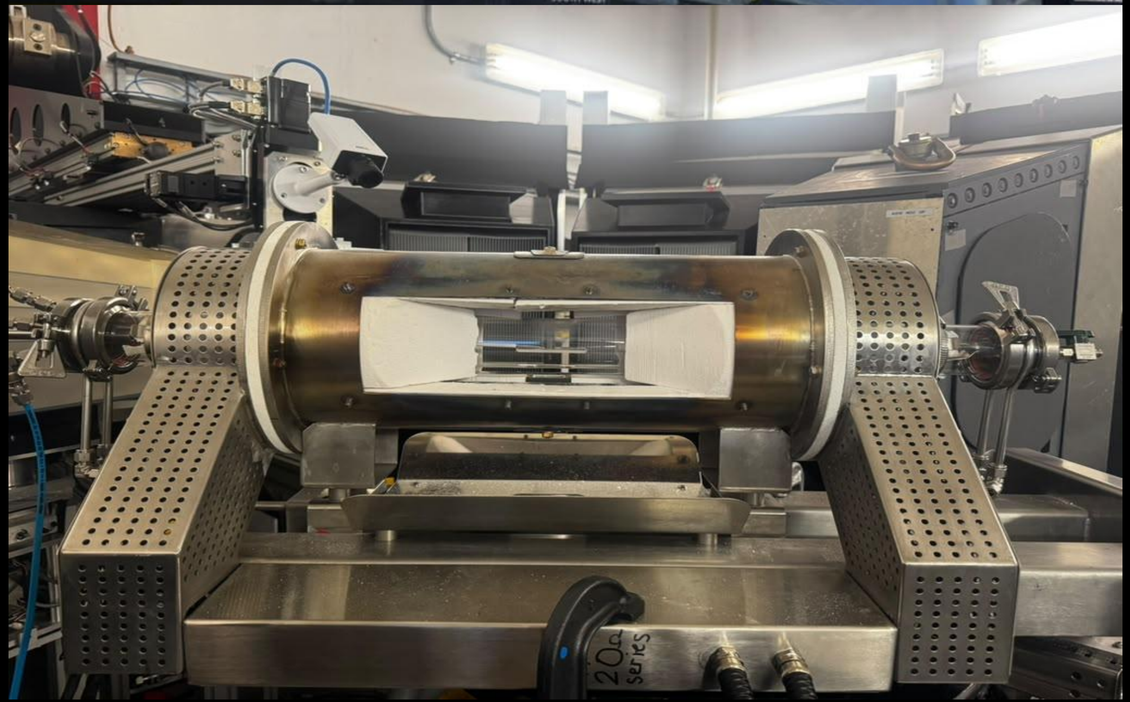
- For Ultrasonic Fatigue Tests, maximizing gain is more important than energy transmission since it is assumed the horn can last millions of cycles
- Horns with identical cylinder lengths and a cosine curve tend to have a higher gain
- Horn length has to be half a wavelength or a full wavelength for the test to work

In-Situ Analysis of Sintered High-Entropy Alloys

Oak Ridge National Laboratory
Argonne National Laboratory
The University of Tennessee Oak Ridge Innovation Institute

Oak Ridge National Laboratory (Vulcan)





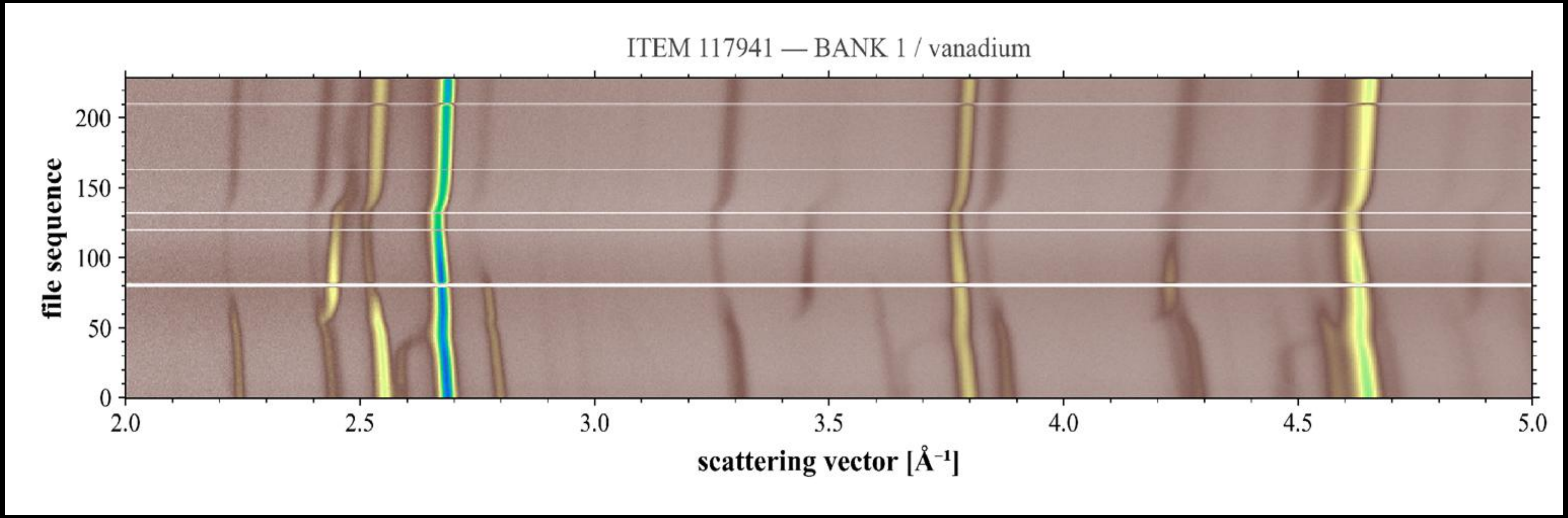
OAK RIDGE National Laboratory

Argonne NATIONAL LABORATORY

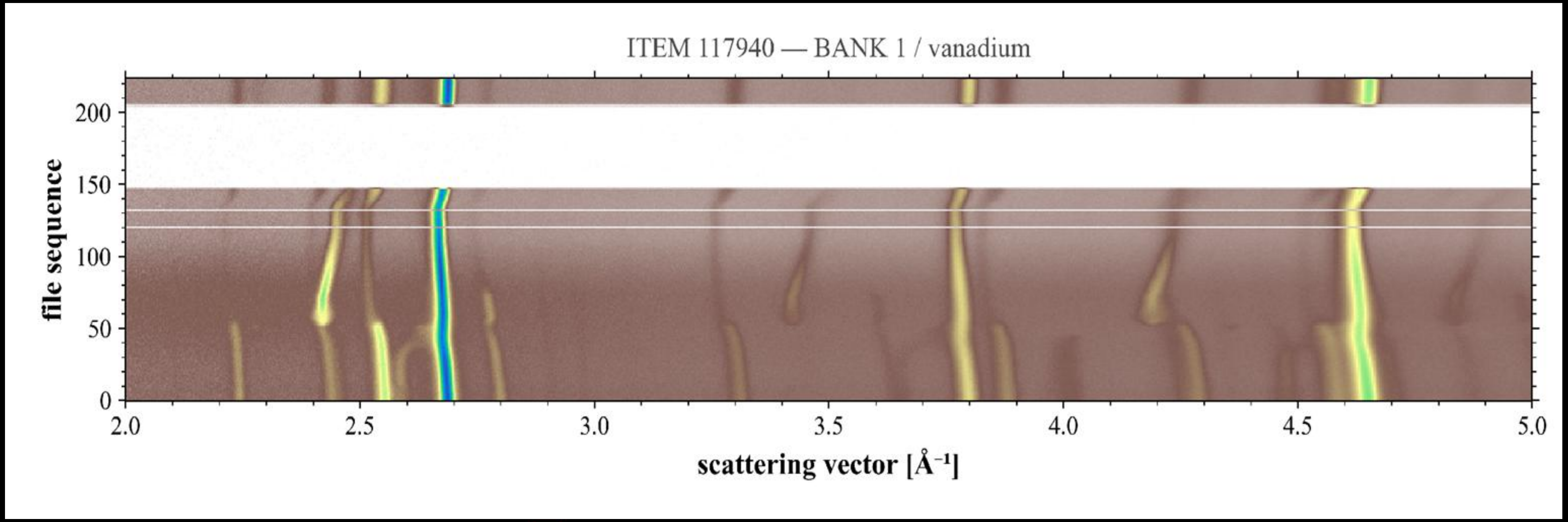
THE UNIVERSITY OF TENNESSEE Oak Ridge Innovation Institute

Neutron Diffraction

Ti-Zr-Nb-Ta

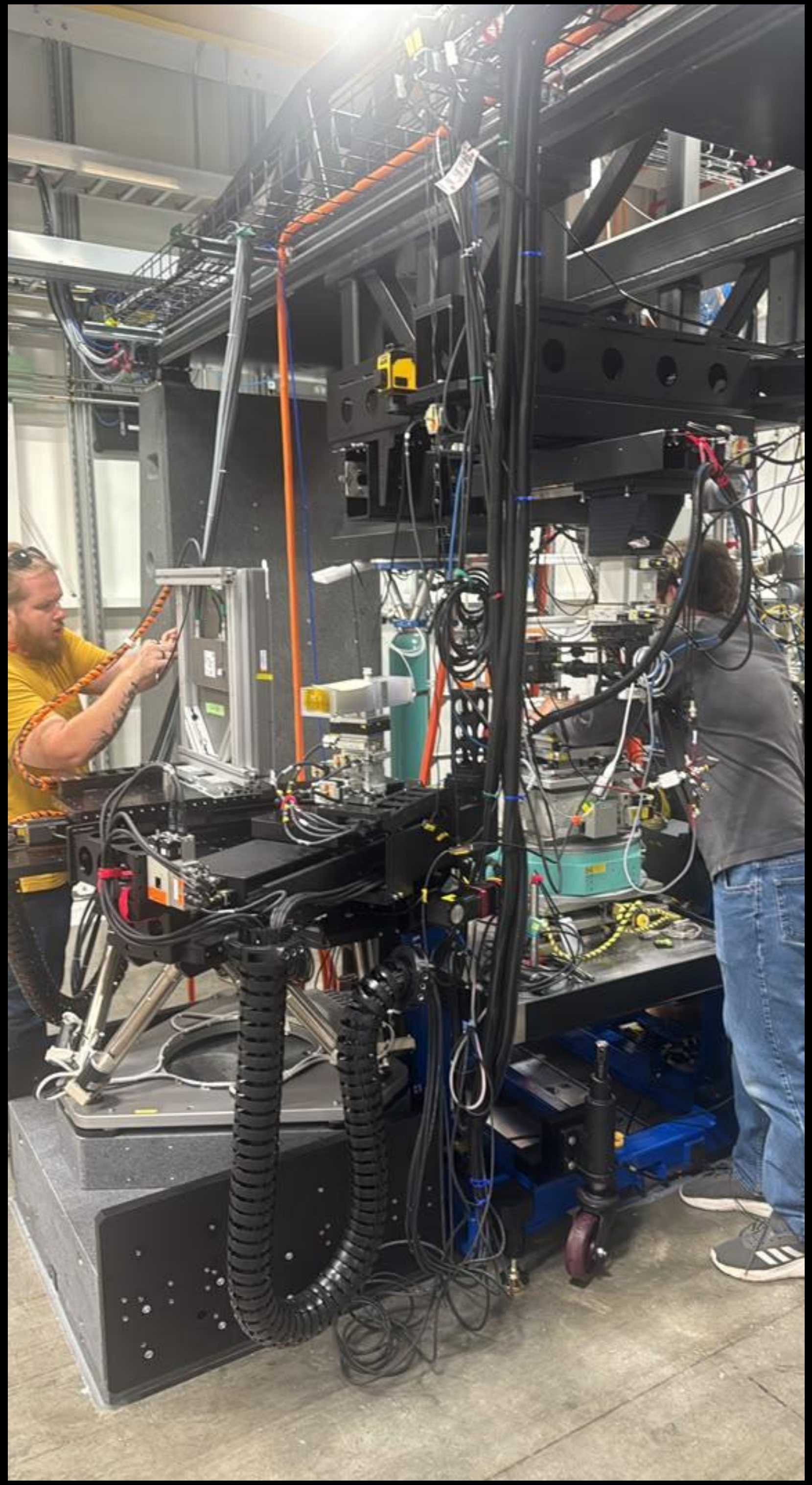
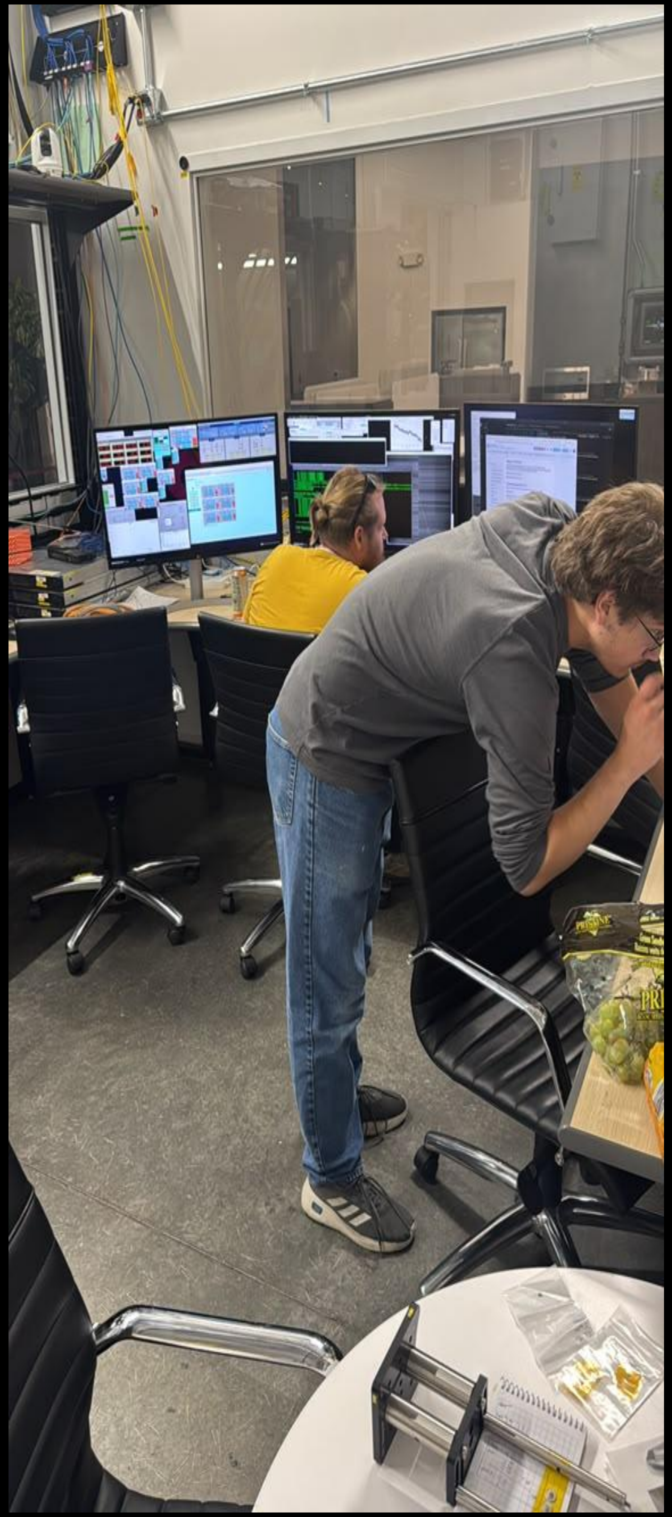
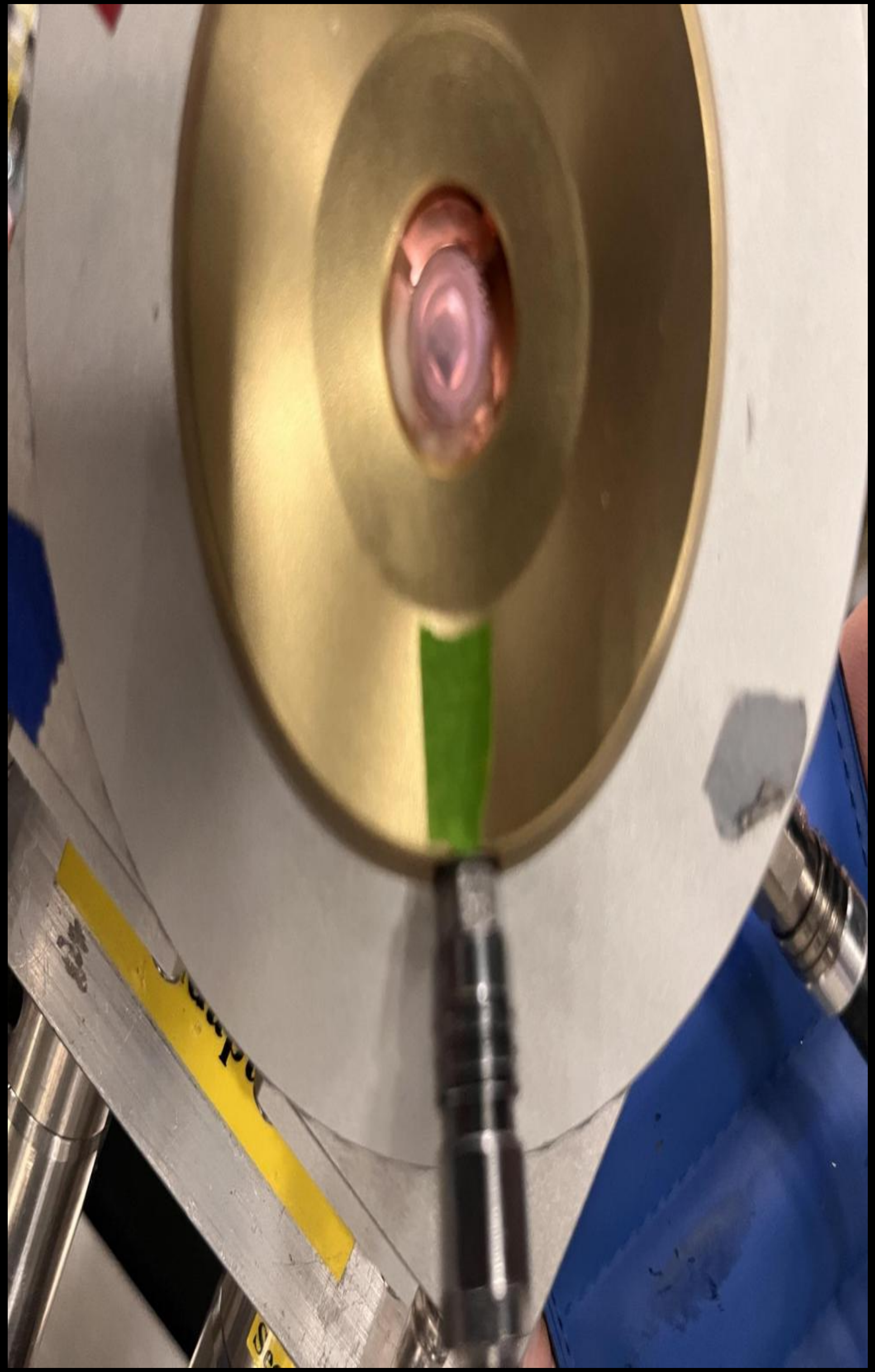


Ti-Zr-Nb-Ta + 10% H



Oak Ridge National Laboratory
Argonne National Laboratory
The University of Tennessee Oak Ridge Innovation Institute

Argonne National Laboratory (HEXM Sector 20)

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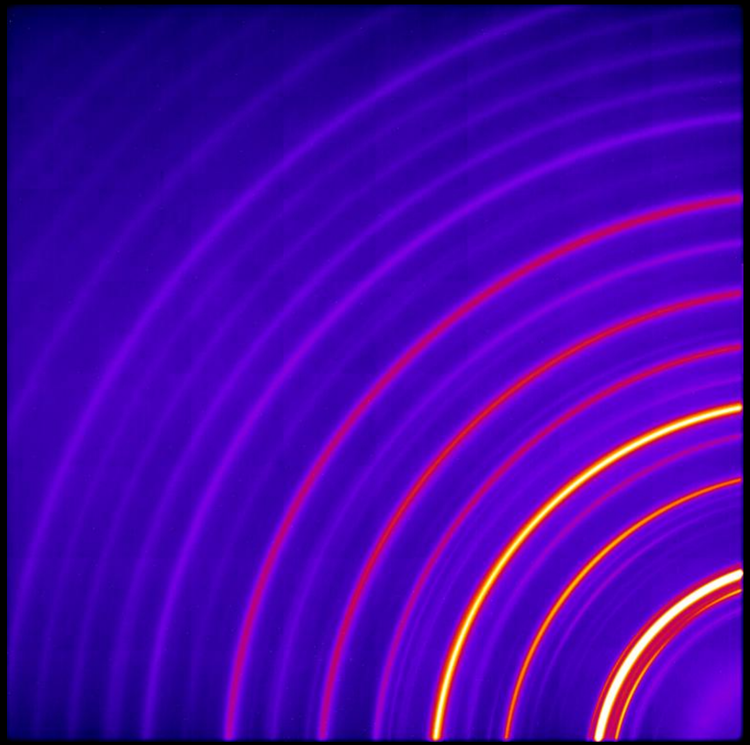
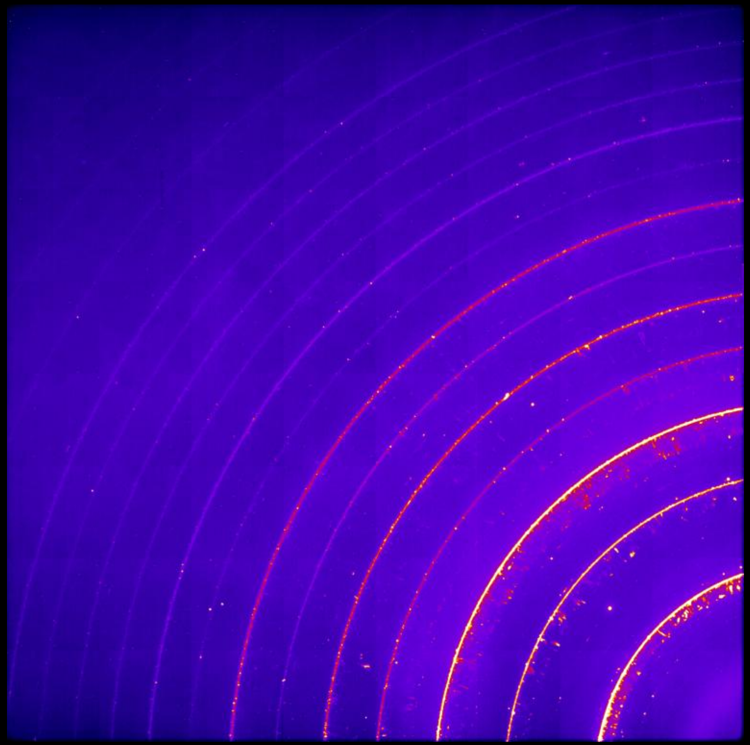
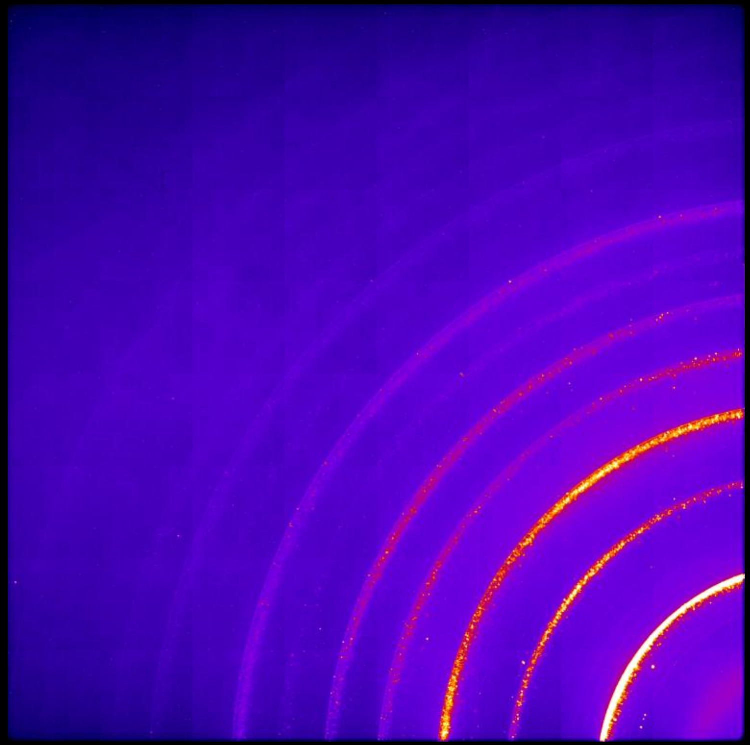
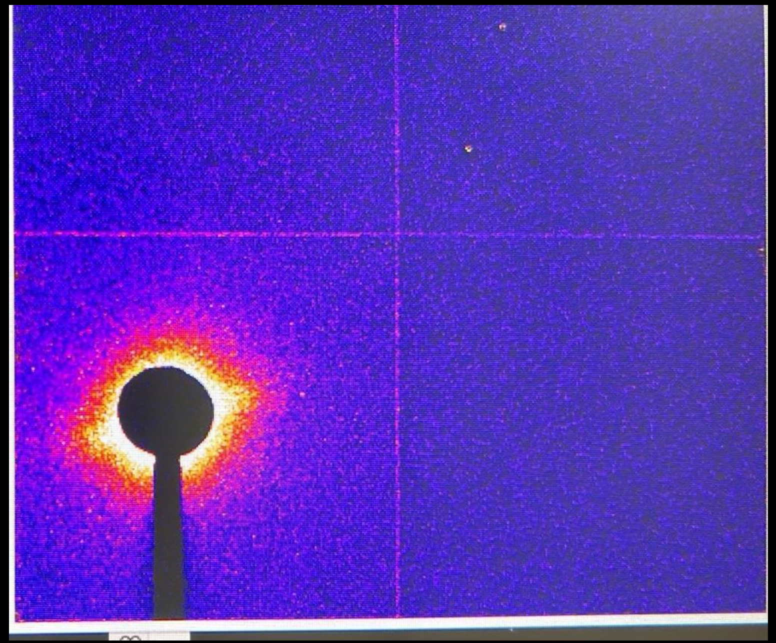
Synchrotron Diffraction


Center at 300 K


Edge at 300 K


Center at 1473 K


Edge at 1473 K


Small angle x-ray

Objective

Oak Ridge National Laboratory:

- Used the Vulcan Beam to see how H_2 would affect the sintering process in high-entropy alloys at the atomic level

Argonne National Laboratory:

- Used the HEXM Beam to see how H_2 would affect the sintering process in high-entropy alloys at the microscopic level

Difference:

- Vulcan Beam uses neutrons, which penetrate deeper into the material
- HEXM Beam uses photons, which don’t penetrate as deeply as neutrons, but provides better resolution at the microscopic scale

Results

- H2 accelerates atom diffusion during sintering
- H2 accelerates grain merging during sintering
- Grains experience strain during sintering, but return to their original shape after cooling down

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Acknowledgements

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Questions?